

Servo Power Requirements

1) MG995 × 3 (Base, Shoulder, Elbow)

Typical MG995 specs (per servo):

- **Stall current:** ~900 mA (0.9 A)
- **Running current (normal load):** ~200–500 mA
- **Idle current:** ~10 mA

For 3 servos:

Condition	Current
Idle	~30 mA
Typical load	0.6–1.5 A
Worst-case (stall) ~2.7 A	

2) SG90 × 2 (Roll, Yaw)

Typical SG90 specs (per servo):

- **Stall current:** ~650 mA
- **Running current:** 100–250 mA
- **Idle:** ~5 mA

For 2 servos:

Condition	Current
Idle	~10 mA
Typical load	0.2–0.5 A
Worst-case (stall) ~1.3 A	

ESP8266 (powered by laptop USB)

Depends on the board (NodeMCU, Wemos D1 Mini).

Typical:

- **Idle WiFi off:** 70–80 mA
- **WiFi active:** 120–200 mA
- **Peak:** 300–400 mA briefly

But **this current is NOT drawn from the servo power supply** if the ESP8266 is powered from your laptop.

However **you must share a common ground** between the servo power supply and ESP.

⚡ Total Power Requirements

Typical load total

MG995 (3×): ~1.5 A

SG90 (2×): ~0.4 A

➡ **~1.9 A typical**

Worst-case (stall)

MG995: ~2.7 A

SG90: ~1.3 A

➡ **~4.0 A worst case**

✅ Recommended Power Supply

To safely power all servos:

Use a 5 V power supply rated for at least 5 A.

(Servos draw high bursts; leaving headroom prevents resets and brownouts.)

Examples:

- 5 V 5 A wall adapter
- 5 V 10 A switching supply (for robotics arms)

Do NOT use the ESP8266's 5 V pin or USB power to drive servos.

🔧 Wiring Tips

- ESP8266 + servo power supplies **must share ground**.
- Power servos directly from the **5 V 5 A supply**, not from the ESP.
- Use a **servo power bus** or a breadboard power module.
- Consider adding a **1000 µF capacitor** across the servo 5 V rails to smooth current spikes.

■ Summary

Component	Current (typical)	Current (max)
3× MG995	0.6–1.5 A	~2.7 A
2× SG90	0.2–0.5 A	~1.3 A
Total (servos)	1.9 A	~4.0 A
ESP8266 (USB)	120–200 mA	300–400 mA

Use a 5V 5A power supply for the servos.

ESP8266 stays powered separately (USB), but grounds must be common.